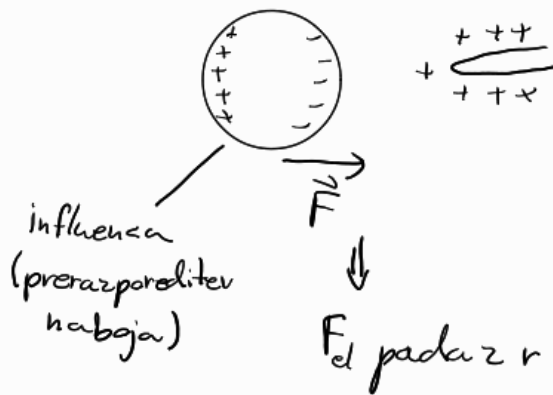


naboj $\leftrightarrow \oplus$; nevtralen $n_+ = n_-$

↳ gibljivi - prevodnik

↳ fiksni - izolator

e, q [As]
američani



$$F_g \leftrightarrow F_{el}$$

$$\downarrow \qquad \downarrow$$

$$\frac{\kappa m_1 m_2}{r^2} \left(-\frac{\vec{r}}{r} \right) \qquad \boxed{\frac{e_1 e_2}{4\pi \epsilon_0 r^2} \left(\frac{\vec{r}}{r} \right)}$$

Coulombov zakon

ϵ_0 - električna/influenčna konstanta

$$\epsilon_0 = \frac{1}{4\pi \cdot 9 \cdot 10^9} \frac{(As)^2}{Nm^2}$$

osnovni naboj

$$e_0 = 1,6 \cdot 10^{-19} As$$



$$v \ 1kg \ H \rightarrow 10^6 As$$

$$e_0^+ = e_0^- \quad m_p = 1836 m_e$$

$$p_0^+$$

 $m_p = \frac{1kg}{N_A}$

$$e_0^-$$

 $m_e = \frac{m_p}{1836}$

$$F_e = \frac{(1,6 \cdot 10^{-19})^2 \cdot 4\pi \cdot 9 \cdot 10^9 N}{4\pi \cdot 10^{-20}} = \underline{3 \cdot 10^{-8} N}$$

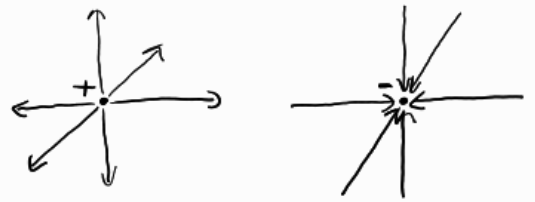
$$F_g = \frac{6,6 \cdot 10^{-11}}{1836 \cdot (6 \cdot 10^{26})^2 \cdot 10^{-20}} = \underline{10^{-46} N}$$

El. polje

$$\vec{F}_{zel} = \vec{E}_1 e_2 \rightarrow \vec{E} = \frac{e}{4\pi\epsilon_0 r^2} \left(\frac{\vec{r}}{r} \right)$$

(točkast naboj)

silnice el. polja



tangenta $\rightarrow$ smer polja
gestota $\rightarrow$ velikost polja

Porazdelitev naboja

- po volumnu

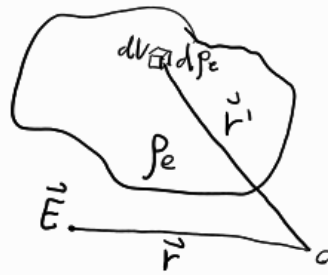
$$\rho_e = \frac{de}{dV}$$

- po površini (votla krogla)

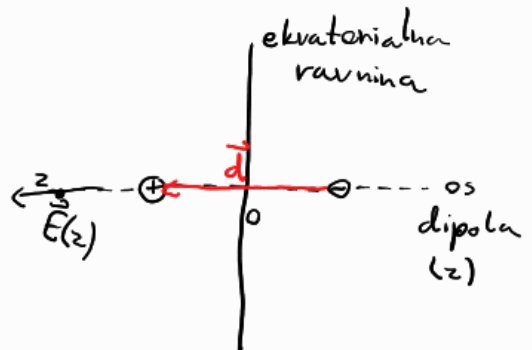
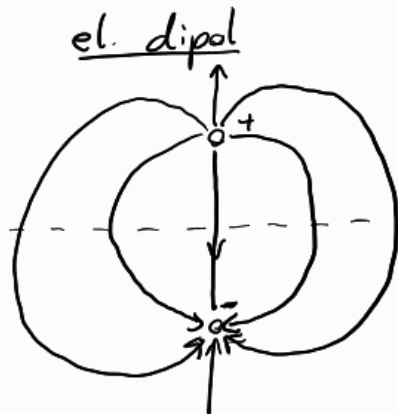
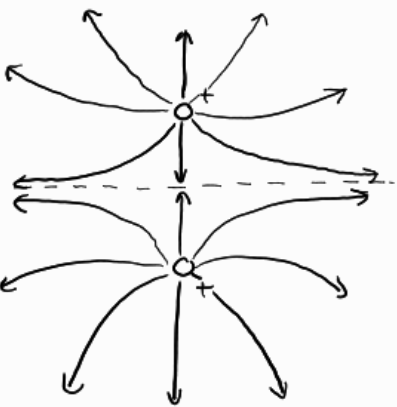
$$\sigma = \frac{de}{dS}$$

- po dolžini (dolga žica)

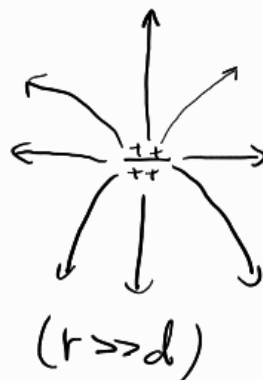
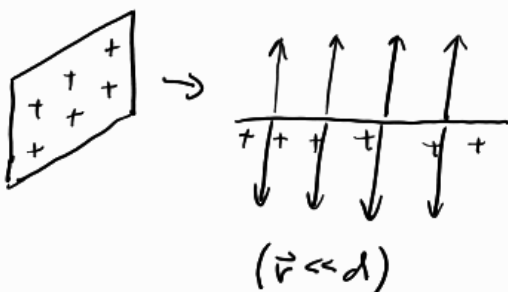
$$\lambda = \frac{de}{dl}$$



$$\vec{E}(\vec{r}) = \int d\vec{E} = \int \frac{d\rho(\vec{r}') dV}{4\pi\epsilon_0 (\vec{r} - \vec{r}')^2} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|}$$



Homogeno polje

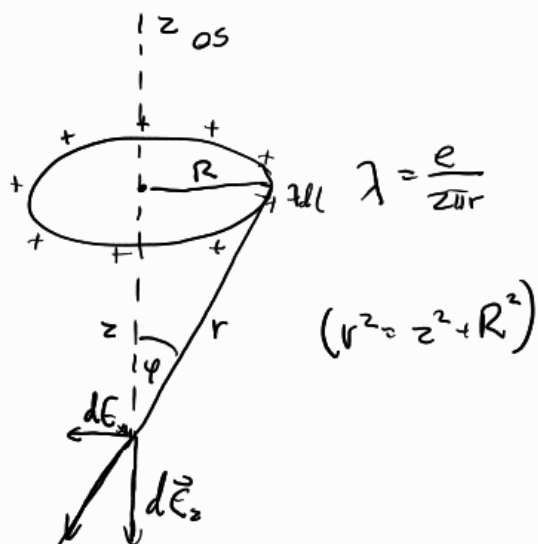


$$\begin{aligned} \vec{E}_z(z) &= E_+ - E_- = \frac{e}{4\pi\epsilon_0} \left(\frac{1}{r_+^2} - \frac{1}{r_-^2} \right) \\ &= \frac{e}{4\pi\epsilon_0} \left(\frac{1}{(z - \frac{d}{2})^2} - \frac{1}{(z + \frac{d}{2})^2} \right) \\ &= \frac{e}{4\pi\epsilon_0} \left(\frac{2zd}{(z^2 - \frac{d^2}{4})^2} \right) \stackrel{z \gg d}{=} \\ &= \frac{2ed}{4\pi\epsilon_0 z^3} \end{aligned}$$

električni dipol $\vec{p}_e = ed$

velja za celoten prostor na veliki razdalji
hi dipola $\rightarrow$ "neutralno"

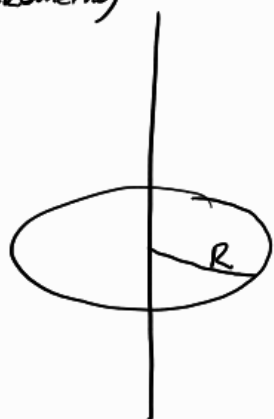
Nabitá kružná zanka



$$\int dE_{xy} = 0$$

$$\int dE_z = \int \frac{\lambda R d\varphi}{4\pi\epsilon_0 r^2} \cdot \frac{z}{r} = \frac{e \cdot z}{4\pi\epsilon_0 r^3} = \frac{e}{4\pi\epsilon_0 (R^2 + z^2)} \frac{z}{\sqrt{z^2 + R^2}}$$

Nabitá kružná plošča (enakomerno)



$$de = \sigma dS = \frac{e}{\pi R^2} \cdot 2\pi r dr$$

$$E_z = \int_0^R \frac{de \cdot z}{4\pi\epsilon_0 (r^2 + z^2)^{3/2}} \cdot \frac{1}{R^2} = \frac{e z}{4\pi\epsilon_0 R^2} \int_0^R \frac{2r dr}{(r^2 + z^2)^{3/2}} = \frac{e z}{4\pi\epsilon_0 R^2} \int_{z^2}^{R^2 + z^2} \frac{du}{u^{3/2}} =$$

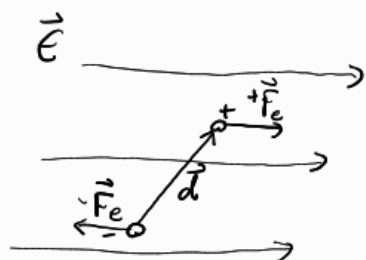
$$= \frac{e z}{4\pi\epsilon_0 R^2} (-2) \left(\frac{1}{\sqrt{R^2 + z^2}} - \frac{1}{z} \right) =$$

$$= \frac{e z}{2\pi\epsilon_0 R^2} \left(\frac{1}{z} - \frac{1}{\sqrt{R^2 + z^2}} \right) = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{R^2 + z^2}} \right)$$

$$u = r^2 + z^2$$

$$du = 2r dr$$

Dipol v el. polju



homogeno $\rightarrow F = 0$

$$\vec{M} = \vec{r} \times \vec{F} = \vec{d} \times e \vec{E} = \boxed{\vec{p}_e \times \vec{E}}$$

$$A = \int \vec{M} \cdot d\vec{\varphi} = \int p_e \sin\varphi d\varphi = -p_e \cos\varphi = \boxed{-\vec{p}_e \cdot \vec{E} = W_{pe}}$$

Gaussov zakon

$$\Phi_v = \int \vec{v} \cdot d\vec{S}$$

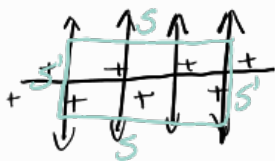
↓

$$\Phi_e = \epsilon_0 \int \vec{E} \cdot d\vec{S} \quad \left. \begin{array}{l} \text{šteti je sili} \\ \text{integral po zaklj. ploskvi je} \\ \text{enak naboju znotraj ploskve} \end{array} \right\}$$

$$e = \epsilon_0 \oint \vec{E} \cdot d\vec{S} \quad \text{Gaussov zakon}$$

↓ izvemo samo, kakšen je prebitok +/- nabojev, ne pa celotno število nabitih delcev

Plošča:



$$\Phi_e(s') = 0$$

$$\Phi_e = \epsilon_0 E \cdot 2S$$

$$e = \epsilon_0 E \cdot 2S$$

$$\frac{e}{S} = 2\epsilon_0 E$$

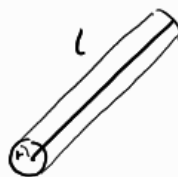
Coulomb ↔ Gauss



$$\epsilon_0 \oint \vec{E} \cdot d\vec{S} = \epsilon_0 E(r) \cdot 4\pi r^2 = \frac{e \cdot \cancel{\epsilon_0} \cdot \cancel{4\pi} r^2}{\cancel{4\pi} \epsilon_0 r^2} = e$$

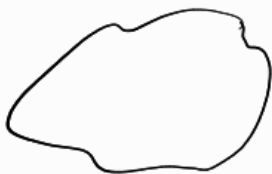
$\vec{r} < a$:

$$\Phi_e = \epsilon_0 E(r) S = \epsilon_0 E(r) \cdot 2\pi r l = e$$



$$E(r) = \frac{\lambda}{2\pi \epsilon_0 r}$$

v prevodniku:



naboji na miru - el. statika

↓

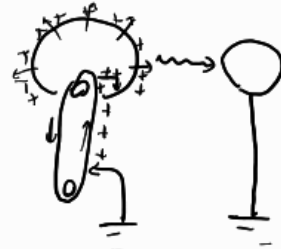
$$\vec{E}_{not} = 0 \xrightarrow{\text{Gauss}} e_{int} = 0 \rightarrow e_+ = |e_-|$$

↓



rotlina → Faradayeva kletka

Van de Graaf generator:



El. pot. en.

$$\Delta W_{ep}(\vec{r}, \vec{r}') = -A_e = -\int_{\vec{r}'}^{\vec{r}} \vec{F}_e \cdot d\vec{s}$$

$$\vec{F}_e = \frac{e_1 e_2}{4\pi\epsilon_0 r^2} \frac{\vec{r}}{r}$$

$$\vec{F}_g = -\frac{\chi_{m_1 m_2}}{r^2} \frac{\vec{r}}{r}$$

$$\Downarrow$$

$$W_{p,e} = \frac{e_1 e_2}{4\pi\epsilon_0 r}$$

$$\Downarrow$$

$$W_{p,g} = -\frac{\chi_{m_1 m_2}}{r}$$

$$F_e = eE \Rightarrow W_{p,e} = eV(\vec{r}) \Rightarrow \Delta W_{p,e} = eU(\vec{r}, \vec{r}')$$

$\uparrow$ el. potencial $\uparrow$ el. napetost

$$U(\vec{r}, \vec{r}') = -\frac{A_e}{e} = -\frac{\int_{\vec{r}'}^{\vec{r}} \vec{F}_e \cdot d\vec{s}}{e} = -\int_{\vec{r}'}^{\vec{r}} \vec{E} \cdot d\vec{s} \quad [V = \frac{N_{lm}}{A_s}]$$

$$V(\vec{r}) = U(\vec{r}, \infty)$$

$$V(\vec{r}) = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{s}$$

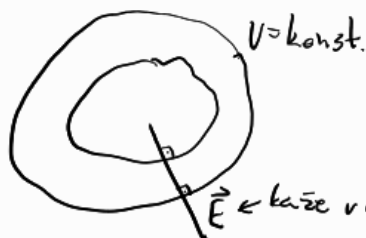
$$\downarrow$$

$$\sum_0 \left[\frac{(A_s)^2}{N_{lm}^2} = \frac{A_s}{V_{lm}} \right]$$

$$E \left[\frac{N}{A_s} = \frac{V}{m} \right]$$

vekt. polje $\vec{E} \rightarrow$ skalarno polje $V(\vec{r})$

$\downarrow$
ekvipotencialne ploskve



$\vec{E}$ kaže v nasp. strani večanja (če je notranji V večji od zunanjega, $\vec{E}$ kaže ven)

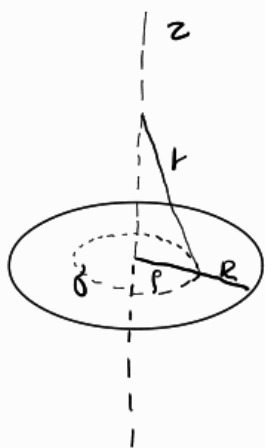
$$\vec{E} = \sum \vec{E}_i \Rightarrow V = \sum V_i = \sum \frac{e_i}{4\pi\epsilon_0 |\vec{r} - \vec{r}_i|} \rightarrow \text{za zvezno porazdeljen naboj}$$

za točk. naboje:

$$V(\vec{r}) = \frac{e}{4\pi\epsilon_0 r}$$

$$V = \int \frac{dV}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|}$$

el. pot. na z-osi enakomerno
rabite plošče

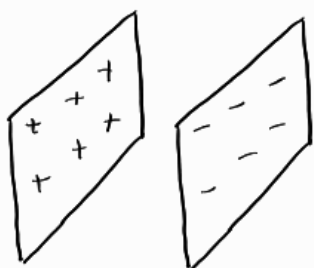


$$V(r) = \int \frac{dq}{4\pi\epsilon_0 r} = \frac{q}{s} \int_0^R \frac{2\pi\rho d\rho}{4\pi\epsilon_0 \sqrt{z^2 + \rho^2}} = \frac{q \cdot z}{4\pi\epsilon_0 R^2 (\sqrt{z^2 + R^2} - z)}$$

$$= \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - z)$$

Shranba $W_e \rightarrow$ kondenzator

2 plošči :



$$e = CU$$

kapaciteta

$$\left[\frac{As}{V} = F \right]$$

Farad



$$E = \frac{\sigma}{\epsilon_0} \rightarrow U = Ed = \frac{q}{\epsilon_0 S} d$$

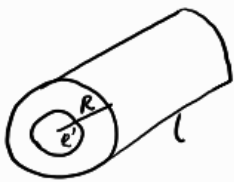
$$e = \left[\frac{\epsilon_0 S}{d} \right] U$$

$$C = \frac{\epsilon_0 S}{d}$$

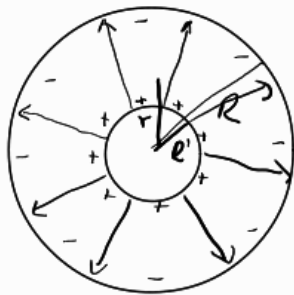
tipično nF, pF

v žipih fF

femto



valjast kondenzator



E ni homogeno

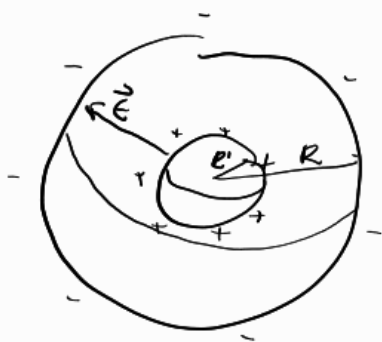
$$E(r) = \frac{e/l}{2\pi\epsilon_0 r}$$

$$U = - \int_{R'}^R E(r) dr = \frac{e/l}{2\pi\epsilon_0} \ln \frac{R}{R'}$$

||

$$C = \frac{2\pi\epsilon_0 l}{\ln \frac{R}{R'}}$$

krogelni kondenzator



$$E(r) = \frac{e}{4\pi r^2 \epsilon_0}$$

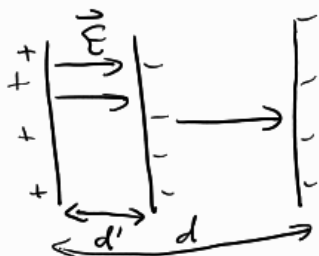
$$U = \frac{e}{4\pi\epsilon_0} \left(\frac{1}{R'} - \frac{1}{R} \right) =$$

$$= \frac{e}{4\pi\epsilon_0} \left(\frac{R-R'}{RR'} \right)$$

$$\Rightarrow C = \frac{4\pi\epsilon_0 RR'}{R-R'}$$

→ če gre $R \rightarrow \infty$: $C = 4\pi\epsilon_0 R'$

Delo kondenzatorja



$$A = \int F ds = F(d-d')$$

$$F = e_2 E_1 = e \frac{E}{2}$$

$$A = e \frac{E}{2} (d-d') = \frac{e(U-U')}{2} \rightarrow dW_{ep}$$

$$W_{ep} = 0 \Leftrightarrow d = 0$$

$$W_c = \frac{eU}{2} = \frac{CU^2}{2} = \frac{e^2}{2C} > 0$$

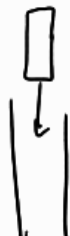
energija kondenzatorja

$dA = C U dU = \underline{U de}$

$$w_c = \frac{W_c}{V} = \frac{\epsilon_0 E^2}{2} \approx \frac{\epsilon_0 E^2}{2}$$

lastnost (praznega!) prostora

Snov v el. polju



$$C_0 = \frac{\epsilon_0 S}{d}$$

$$C > C_0$$

ko v kond. vstavimo
neko snov

$$C = \epsilon C_0$$

dielektričnost

$$C \rightarrow C_0 \epsilon$$

$$\epsilon \rightarrow \epsilon_0 \epsilon$$

$$U = \frac{e}{C} = \frac{e_0 \phi}{C_0 \phi}$$

~~$$E = \frac{U}{d} = E_0$$~~

$$2) f = \frac{e}{\epsilon_0 S} = \epsilon E_0$$

$$\boxed{\epsilon_0 \rightarrow \epsilon_0 \epsilon}$$

$$C = \frac{\epsilon \epsilon_0 S}{d}$$

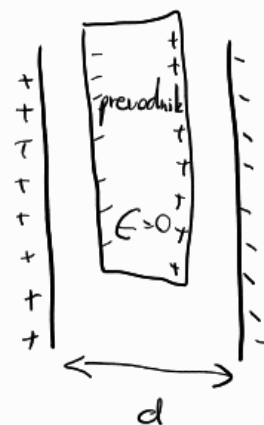
$$E = \frac{e}{4\pi \epsilon_0 \epsilon r^2}$$

$$\phi_e = \epsilon \epsilon_0 \int \vec{E} \cdot d\vec{S}$$

$$e = \epsilon \epsilon_0 \oint \vec{E} d\vec{S}$$

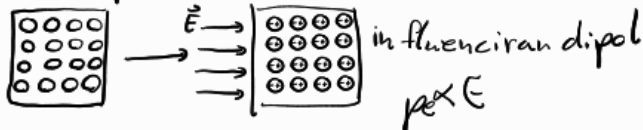
$$w_c = \frac{\epsilon \epsilon_0 E^2}{2}$$

snov	ϵ
plastika	3-5
zrak	~ 1 (zares 1,00054)
H ₂ O	~ 80
keramike	$\sim 500-1000$

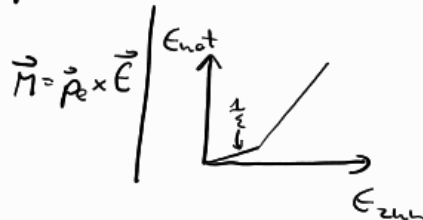
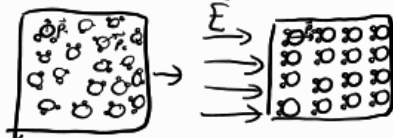


vezan naboj
SENČENJE
NABOJA

Nepolarne molekule



Polarne molekule

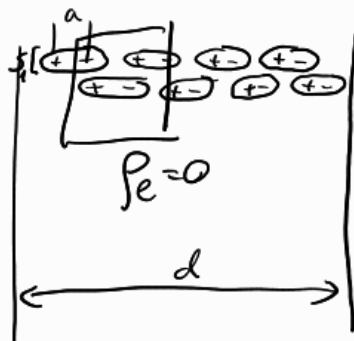


$$e^{-\frac{\Delta W_p}{k_B T}} = e^{-\frac{2p_e E}{k_B T}} \propto E \Rightarrow \boxed{K p_e \propto E}$$

za vodo

$$\Delta W_p = 2p_e E \approx 2 \cdot 30 e_0 pm \cdot 1 \frac{MV}{m} \approx 10^{-4} eV$$

$$k_B T = \frac{1}{40} eV$$



$$S = N_s \cdot S_1$$

$$N = N_s \cdot N_d \leftarrow \begin{array}{l} \text{št. molekul v verigi} \\ \text{št. verig} \end{array}$$

$$\sigma_{\text{vezan}} = \frac{q_{\text{vezan}}}{S_1} = \frac{e_0}{S_1} = \frac{e_0 N_s N}{S N_d d} = \frac{N \langle \vec{p}_e \rangle}{V} = \vec{P}$$

$$N_d = \frac{d}{a}$$

$\vec{P}$
polarizacija
("gostota dipolov")

$$\epsilon_0 \vec{E}_{\text{NOT}} = \vec{E}_{\text{znn}} - \vec{P}$$

$$\vec{E}_{\text{NOT}} = \frac{\vec{E}_{\text{znn}}}{\epsilon}$$

$$\epsilon \gtrless 1$$

↑
snov
↑
molekul

keramike

↳ domene (skupki z isto usmerjenimi dipoli)

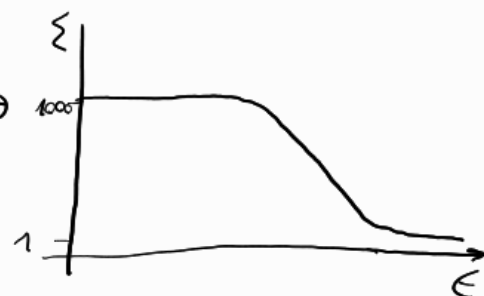
$$\left(\uparrow \uparrow \uparrow \uparrow \right) p_0 = N p_1$$

$> 10^3$

$$p_0 \gg p_e \rightarrow p_0 \neq E \rightarrow$$

(feroelektrik)

↓
pri majhnih E je
 ϵ visok



$$\left. \begin{array}{l} \vec{P} \neq \vec{E} \\ \vec{P} \neq \vec{E} \end{array} \right\} \text{anizotropni dielektrik}$$

$\vec{E} \rightarrow$ jakost električnega polja

$$\vec{F} = e \vec{E}$$

$$V = - \int \vec{E} \cdot d\vec{s}$$

$\vec{D} \rightarrow$ gostota električnega polja

$$\Phi_e = \int \vec{D} \cdot d\vec{S} = \epsilon_0 \int \vec{E} \cdot d\vec{S}$$

↓
vakuum: $\vec{D} = \epsilon_0 \vec{E}$

snov: $\vec{D} = \epsilon \epsilon_0 \vec{E} = \epsilon_0 \vec{E} + \vec{P}$

$$\Rightarrow \text{Gauss: } \oint \vec{D} \cdot d\vec{S} = q$$

↑
anizotropna
snov
 $\Rightarrow \epsilon$ postane
tenzor

